

EARN - project status report

Coverage-aware, ledger-anchored federated aggregation for rare skin disease diagnosis

Pipeline & EARN track (Ananya) | 20 September 2026 | Week 1 of 6

Repository: github.com/ananyac9820/fedrare | Live site: fedrare.vercel.app

All figures in this report are computed from the repository's own result files (last synced 2026-09-19). Nothing is estimated.

1. Summary

The data work and the whole Tier A software pipeline are finished and tested, and the project's central measured finding is confirmed: on the real six-hospital split, the hospital holding most of the rare-disease data (Centre 2) receives far less weight than its knowledge under standard federated averaging - a **1.87x** mismatch. A public site presenting all of this is deployed.

Two pre-registered checks were run and **both failed**: the evidence signal EARN depends on (G0a) and the quality of the Tier A baseline (G0b). Neither failure was worked around or re-tuned. Section 5 turns each failure into a specific change with an owner; those changes are now the critical path, because EARN itself cannot be built until G0a is fixed.

Nothing about EARN has been validated. No trust score, attack result or ledger measurement exists yet, and none is shown anywhere in the project.

Area	State	Evidence / note
Dataset analysis and the problem claim	Done	23,247 images, 6 hospitals, 8 diseases; 1.87x weight-vs-knowledge mismatch
Feature extraction (Tier A)	Done	All images converted to 1024-d DenseNet-121 features; checked against torchvision
Federated loop, aggregation interface, baselines	Done	FedAvg, Krum, multi-Krum, trimmed mean, median, Camp A; 30 unit tests pass
Constructed split S2	Done	Rare-class coverage drops to 1 holder; fixed seed
FedAvg baseline on the natural split S1	Done	3 seeds x 40 rounds; balanced accuracy 0.415
Gate G0a - evidence signal	FAILED	Mean Spearman -0.21, needed at least 0.7
Gate G0b - baseline quality	FAILED	Balanced accuracy 0.415, needed at least 0.45
Camp A baseline	Blocked	Depends on the G0a fix
EARN aggregation rule	Not run	Week 3; only if G0a is fixed
Attacks A1-A3 and Gate G1	Not run	Arya, Week 2
Smart contract, gas and latency	Not run	Aditya, Week 5
Public status site	Deployed	fedrare.vercel.app , rebuilt on every push

2. What has been built

All of the following is committed and pushed. Branch `feature/pipeline` holds the research code; `main` additionally holds the website.

Component	Where	What it does
Data loading	src/data/loader.py	Fed-ISIC2019 from the Hugging Face mirror; reads label/centre columns without decoding images
Distribution analysis	scripts/02_explore_data.py	Per-hospital, per-disease counts; rare-class rule; over-representation per hospital
Feature extraction	scripts/05_extract_features.py	Frozen ImageNet DenseNet-121 over every image once, saved for reuse (Tier A)
Aggregation interface	src/federated/interface.py, docs/aggregation_interface.md	The one contract shared with the attacks track; validated every round
Aggregation rules	src/federated/baselines.py, robust.py, primitives.py	FedAvg, FedAvg+clipping, Krum, multi-Krum, trimmed mean, median, Camp A
Tier A training loop	src/federated/tier_a.py	Trains only the classifier head on saved features; hooks let attacks replace a client
Splits S1 / S2	src/data/splits.py	Natural split and the constructed specialist split, fixed seed so everyone gets the same S2
Gate scripts	scripts/06_gate_g0a.py, 07_tier_a_s1_baselines.py	Run the two Week-1 checks and write their records to results/
Tests	tests/	30 unit tests covering the rules, primitives, loop and splits
Website	web/	Six-page Next.js site reading only the result files; deployed on Vercel

3. Measured results (verified)

Every number below comes from the project's own runs on real data.

3.1 The dataset

Property	Value
Dataset	Fed-ISIC2019 (FLamby-derived Hugging Face mirror)
Images	23,247 total; 18,597 training, 4,650 test
Hospitals	6 real data centres (not simulated splits)
Diseases	8 classes
Rare classes	Dermatofibroma 1.03% and Vascular lesion 1.09% of all images
Rare-class rule	Under 1/25 of the largest class (Melanocytic nevus, 48.72%), i.e. a 1.95% cutoff. Chosen after seeing the counts; disclosed as such.

3.2 The headline finding: knowledge does not match weight

Under FedAvg a hospital's influence equals its share of all training images. Centre 2 holds 27.0% of the rare-disease training images but receives 14.5% of the weight - a 1.87x gap. No other hospital shows a comparable gap.

Hospital	Images	Training images	Rare-disease training images	Share of rare data	FedAvg weight	Ratio
Centre 0	12,413	9,930	184	47.8%	53.4%	0.90x
Centre 1	3,954	3,163	68	17.7%	17.0%	1.04x
Centre 2 (specialist)	3,363	2,691	104	27.0%	14.5%	1.87x
Centre 3	2,259	1,807	23	6.0%	9.7%	0.61x
Centre 4 (holds none)	819	655	0	0.0%	3.5%	0.00x
Centre 5	439	351	6	1.6%	1.9%	0.83x

3.3 How many hospitals can vouch for each disease

Counting hospitals with at least 20 training images of a disease. This is the quantity EARN's coverage blend is built on.

Disease	Share of data	Hospitals holding it
Melanoma	18.00%	6 of 6
Melanocytic nevus	48.72%	6 of 6
Basal cell carcinoma	14.29%	3 of 6
Actinic keratosis	3.73%	3 of 6
Benign keratosis	10.44%	5 of 6
Dermatofibroma (rare)	1.03%	4 of 6
Vascular lesion (rare)	1.09%	3 of 6

Disease	Share of data	Hospitals holding it
Squamous cell carcinoma	2.70%	2 of 6

Correction to the design document: it states that common diseases are held by 5 or 6 hospitals. On the real counts that is wrong for three of them - basal cell carcinoma and actinic keratosis have 3 holders, and squamous cell carcinoma has 2. Low coverage is therefore not unique to the rare classes, which strengthens the motivation but must be corrected in the paper.

3.4 FedAvg baseline on the natural split (S1)

Tier A: the classifier head trained on frozen DenseNet-121 features, all six hospitals, 40 rounds, 3 seeds (local steps 50, batch 32, learning rate 0.0005, class-balanced loss). This is the baseline the project compares against - it is not the contribution.

Metric	Result (mean over seeds, final round)
Balanced accuracy	0.415 +/- 0.005
Dermatofibroma F1	0.192 +/- 0.012
Vascular lesion F1	0.471 +/- 0.037
Rare-class average F1	0.332
Overall accuracy	0.591
Per seed (balanced accuracy)	seed 42: 0.417, seed 43: 0.420, seed 44: 0.409
Progress over rounds	round 1: 0.150, round 10: 0.322, round 20: 0.367, round 30: 0.397, round 40: 0.415

The run had not converged: balanced accuracy was still climbing at the final round. That matters for the G0b verdict below.

4. Gate results

Gates are pass/fail checks fixed in the design document before any result existed. The rule is that a failed gate is reported as a failure and triggers its listed fallback - it is never re-tuned until it passes. Both gates run so far have failed.

Gate	When	Condition	Measured	Verdict
G0a - evidence signal	Day 2, Week 1	Evidence e(k,c) ranks hospitals' true per-class counts with Spearman at least 0.7 on S1	Mean Spearman -0.21	FAILED
G0b - baseline quality	End of Week 1	Tier A FedAvg reaches balanced accuracy of at least 0.45	0.415 +/- 0.005 over 3 seeds	FAILED
G1 - problem is real	End of Week 2 (30 Sep)	Under Camp A, attack A1 gives the attacker at least 2x its FedAvg weight on a rare row and drops that class's F1 by at least 0.15, with balanced accuracy falling less than 0.03	-	Pending
G2 - EARN works	End of Week 4 (14 Oct)	Under A1 and A2, EARN's rare-class F1 stays within 0.05 of the no-attack value; honest Centre 2 reaches trust of at least 0.8	-	Pending

4.1 G0a - the evidence signal came out inverted

EARN must infer, from each hospital's model update alone, how much of each disease that hospital holds. The check asked whether this 'evidence' ranks hospitals the way their true image counts do. It scored **-0.21** against a required **0.7**: not merely weak, but pointing the wrong way.

Disease	Spearman correlation	Hospital with the highest evidence
Melanoma	+0.37	Centre 4 - holds 166 images of it
Melanocytic nevus	+0.43	Centre 3 - holds 649 images of it
Basal cell carcinoma	-0.43	Centre 1 - holds 2 images of it
Actinic keratosis	-0.70	Centre 5 - holds 0 images of it
Benign keratosis	+0.43	Centre 0 - holds 923 images of it
Dermatofibroma (rare)	-0.49	Centre 4 - holds 0 images of it
Vascular lesion (rare)	-0.60	Centre 4 - holds 0 images of it
Squamous cell carcinoma	-0.70	Centre 5 - holds 0 images of it

For both rare diseases, and for 2 other diseases, the hospital scoring highest holds **zero** images of that disease. The likely cause is that local training pushes down the score of a disease a hospital never sees, and the measure used (the size of the change to that disease's row) cannot tell a rise from a fall.

4.2 G0b - the baseline is below its bar, and had not converged

FedAvg reached **0.415 +/- 0.005** balanced accuracy against a required **0.45**. All three seeds were below the bar (0.417, 0.420, 0.409). Importantly the run was still improving when it stopped: 0.397 at round 30 and 0.415 at round 40, so the shortfall is at least partly an unfinished run rather than a ceiling. This failure does not block the attack track: it concerns baseline quality, not the evidence signal.

5. Changes required by the failures

Each failure below is translated into a concrete change. These are now the critical path: C1 blocks Camp A, EARN, and everything in Weeks 3 and 4.

#	Change	Why	Owner	State
C1	Fix the evidence signal, then re-run G0a	G0a failed. The design document allows one retry: use the bias row instead of the weight row, or average evidence over 3 rounds. Both keep a magnitude, which is the part that is inverted, so a sign-aware measure should be considered - that is a change to the method and must be recorded as such.	Ananya	Decision open
C2	Re-run G0b with an unfrozen last dense block	The design document's pre-registered fallback for G0b. Requires re-extracting features.	Ananya	In progress
C3	Record the 'more rounds' deviation in writing	Training longer was decided after seeing the result and is not part of the pre-registered fallback. It is defensible because the run had not converged, but it must be written down as an amendment rather than applied silently.	Ananya	To write
C4	Run Camp A on S1 once C1 succeeds	Camp A weights hospitals by evidence, so it cannot be meaningful while the evidence signal is inverted. It is the baseline Gate G1 is defined against.	Ananya	Blocked
C5	Decide the fallback if C1 fails a second time	The design document's rule: after one failed retry, drop EARN and move to Fallback F1 - an empirical study of how existing rare-class-aware methods behave on a real hospital split. The measured material in Section 3 would carry that study.	Team	Decision
C6	Correct the holder-count claim in the design document	Three common diseases have 2-3 holders, not the 5-6 the document states (Section 3.3).	Ananya / Arav	To do
C7	Agree the attack-result file format with the attacks track	G1 is owned by Arya and is due at the end of Week 2. The aggregation interface is already agreed and documented; only the result file shape is open.	Ananya + Arya	This week

6. What is left

6.1 Not run yet - no numbers exist for any of these

Work	State	Owner	Where the result will land
G0b retry: more rounds and an unfrozen last dense block	In progress	Ananya (Pipeline & EARN)	results/gate_g0b.json (re-run)
Camp A baseline on S1	Blocked	Ananya (Pipeline & EARN)	results/tier_a_s1_camp_a.csv
EARN aggregation rule	Not run	Ananya (Pipeline & EARN)	src/federated/earn.py
Attacks A1 / A2 / A3 and Gate G1	Not run	Arya (Attacks & Eval)	results/ (G1 attack table - format to agree)
Real ledger rounds (trust-table and history hashes)	Not run	Aditya (Blockchain)	results/ledger_rounds.json
Ledger gas cost and latency per round	Not run	Aditya (Blockchain)	to agree

6.2 The remaining schedule

Week	Dates	Work	Deliverable
1 (current)	17-23 Sep	Extract DenseNet-121 features. Check G0a on Day 2. Build the Tier A federated loop. Run FedAvg and Camp A on S1. Build split S2.	Features saved; baseline numbers; G0a, G0b
2	24-30 Sep	Attacks A1-A3 against FedAvg, robust rules and Camp A on S1 and S2, 3 seeds.	Attack table; G1; S1/S2 framing decision
3	1-7 Oct	Implement EARN: clipping, evidence, coverage blend, slow-up/fast-down trust, capped weighting. Unit tests per step. In-memory ledger stub.	Working src/federated/earn.py
4	8-14 Oct	Full evaluation: all methods x all attacks x S1/S2 x 3 seeds. Ablations. Sensitivity of step size and threshold.	Main results; G2
5	15-21 Oct	Smart contract on a local Hardhat chain: trust table, history hashes, max-step rule. Gas and latency. Tier B confirmation runs.	Contract + overhead numbers; Tier B table
6	22-28 Oct	Figures, 6-8 page draft, updated synopsis, final literature re-check, clean repository.	Draft paper and final report

7. Earlier questions now closed, and corrections

Item	Outcome
Were robust filters rejecting the rare-disease specialist?	Closed - not supported. Krum selected Centre 2 more often than any other hospital (46.7% rejection against 90.7% for the others). The hypothesis was dropped and the scripts marked superseded. No further compute was spent on it.
A bug that invalidated an earlier verdict	Found and fixed. PyTorch's vector_to_parameters made the model parameters views of the global weight vector, so every client update was exactly zero and all rules produced identical models. The earlier 'weak conflict' verdict was void and was withdrawn.
Are the rare classes under 1% of the data?	Nearly - Dermatofibroma is 1.03% and Vascular lesion is 1.09%, both just above 1%. They qualify as rare by the imbalance-ratio rule, not by a 1% threshold. Worth stating precisely in the paper.
Holder counts for common diseases	The design document's claim of 5-6 holders is wrong for three diseases (Section 3.3).

Item	Outcome
Ledger figures on the website	The chain shown is labelled sample data: the hash links are real SHA-256, the block contents are invented. No gas or latency number is shown anywhere.

8. How to reproduce

```
git clone https://github.com/ananyac9820/fedrare && cd fedrare
pip install -r requirements.txt
python scripts/02_explore_data.py      # per-hospital distribution, rare classes
python scripts/05_extract_features.py # DenseNet-121 features (about 35 min on CPU)
python scripts/06_gate_g0a.py          # Gate G0a
python scripts/07_tier_a_s1_baselines.py --no-camp-a # FedAvg and Gate G0b
python -m pytest                       # 30 unit tests
cd web && npm install && npm run dev    # the site, at localhost:3000
```

The site reads only the JSON files in web/data/. After any new run, `npm run sync-data` regenerates them from results/ and the published site updates on the next push - no code changes needed.

9. Limitations to keep stating

- EARN is unvalidated. It is specified in full and partly implemented in interface terms, but no trust score has ever been computed on real data.
- Both Week-1 gates failed. The project is behind its own plan, and the evidence signal - the foundation of the method - does not currently work.
- The baseline is weak and unconverged, so any later comparison against it is provisional until G0b is re-run.
- All results so far are Tier A: a classifier head on frozen features, not a fine-tuned DenseNet-121. Tier B confirmation runs are scheduled for Week 5 and are the first thing to cut.
- The rare-class rule was chosen after seeing the class counts. This is disclosed in the paper and on the site rather than presented as a prior decision.

Report generated 20 September 2026 from the repository's result files. Figures shown here match the published site exactly.